VLSI DESIGN LAB

Contents

- Introduction
- VLSI front end
- VLSI back end
  - Full custom design
  - Semi Custom design
- VLSI full design

Introduction

Course Information

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VLSI front end

VLSI front end design using XILINX TOOL

- Lab 1: Design and simulate of Carry-Look-Ahead adder.
- Lab 2: Design and simulate of flip flops: SR, D, JK

VLSI front end

Scope :

- Theory:
  - block diagram
  - corresponding boolean expressions
- Verilog program
- Testbench program
- Simulation and results

VLSI front end

Guideline:

- AIM: To design and simulate adder and a flip-flop using Xilinx's VIVADO
- Tools: Xilinx VIVADO suite
- Preferred language: Verilog

VLSI back end

- Full custom design

- Lab 3: Design and analysis of CMOS Inverter.
- Lab 4: Design and analysis of CMOS NAND gate.
- Lab 5: Design and analysis of CMOS D- Flip Flop.

- Semi custom design

- Lab 6: Design and analysis of Full Adder.
- Lab 7: Design and analysis of 3 to 8 Decoder.

VLSI back end

Full custom design

- AIM: To design, simulate and verify the operation of logic gates (INV, NAND, Flip-Flop…) using Cadence tool (or Mentor graphic tool)
- Tools used:
  - 1. Virtuoso Tool for Schematic Designs
  - 2. Spectre/LTSPICE Tool for Simulation

Scope:

- Circuit design
- Test bench
- Simulation result

VLSI back end

Semi custom design

- AIM: Using tool for functional simulation, Synthesis and pre-Layout Timing Analysis and Physical Design

Scope:

- Circuit design
- Test bench
- Simulation pre-layout
- Layout design
- Verification layout design
- Simulation post layout
- Data and result

VLSI back end

Semi custom design

- ASIC flow step by step: follow guideline in attached document.